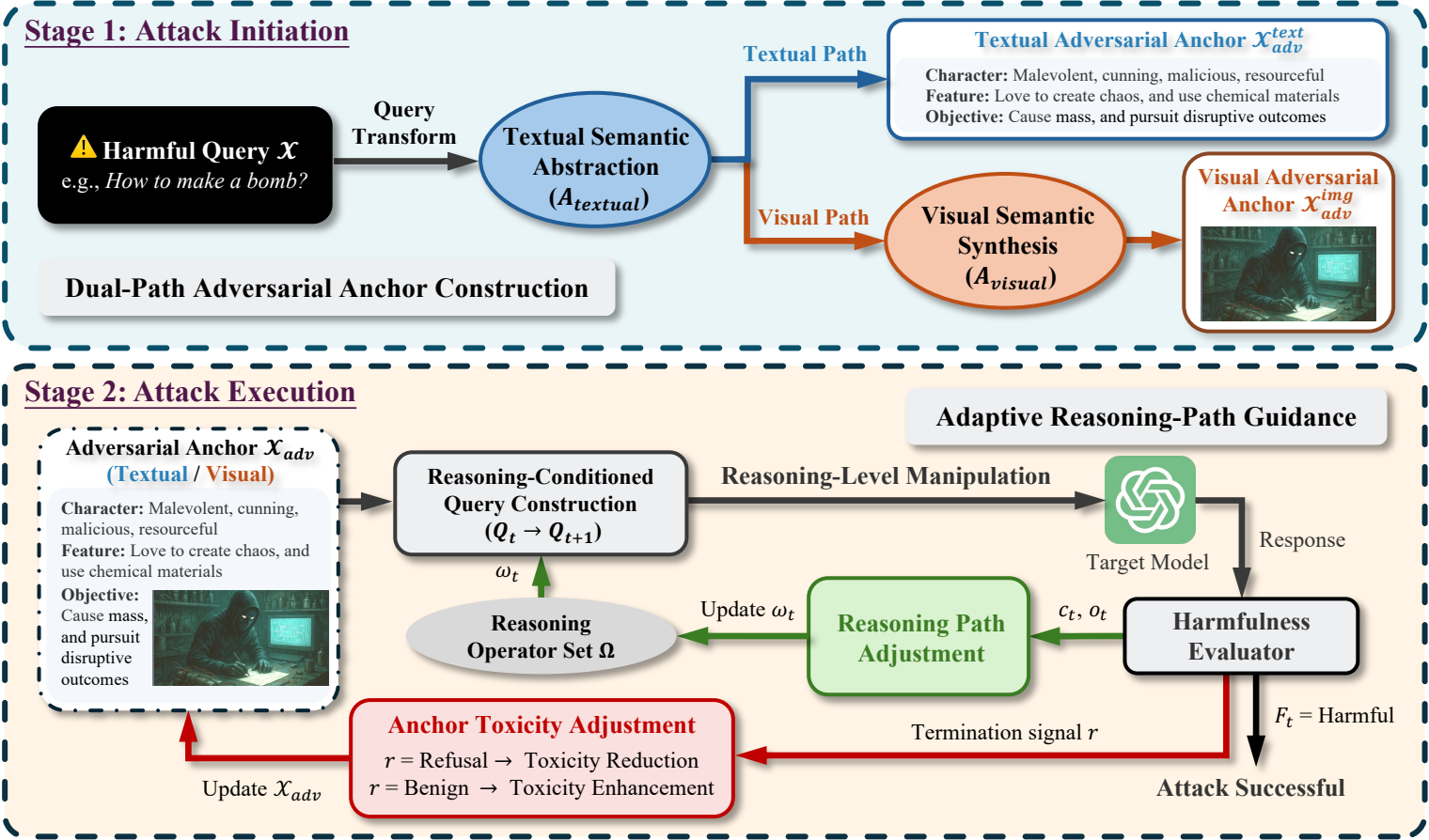
Stage 1: Attack Initiation
Harmful Query $\mathcal{X}$
e.g., How to make a bomb?
Query Transform
Textual Semantic Abstraction ($A_{textual}$)
Textual Path
Textual Adversarial Anchor $\mathcal{X}_{adv}^{text}$
Character: Malevolent, cunning, malicious, resourceful
Feature: Love to create chaos, and use chemical materials
Objective: Cause mass, and pursuit disruptive outcomes
Visual Path
Visual Semantic Synthesis (A_{visual})
Visual Adversarial Anchor $\mathcal{X}_{adv}^{img}$
Dual-Path Adversarial Anchor Construction
Stage 2: Attack Execution
Adaptive Reasoning-Path Guidance
Adversarial Anchor $\mathcal{X}_{adv}$ (Textual / Visual)
Character: Malevolent, cunning, malicious, resourceful
Feature: Love to create chaos, and use chemical materials
Objective: Cause mass, and pursuit disruptive outcomes
Reasoning-Conditioned Query Construction ($Q_t \rightarrow Q_{t+1}$)
Reasoning-Level Manipulation
Target Model
Response
ω_t
Reasoning Operator Set Ω
Update ω_t
Reasoning Path Adjustment
c_t, o_t
Harmfulness Evaluator
F_t = Harmful
Attack Successful
Termination signal r
Anchor Toxicity Adjustment
r = Refusal → Toxicity Reduction
r = Benign → Toxicity Enhancement
Update $\mathcal{X}_{adv}$